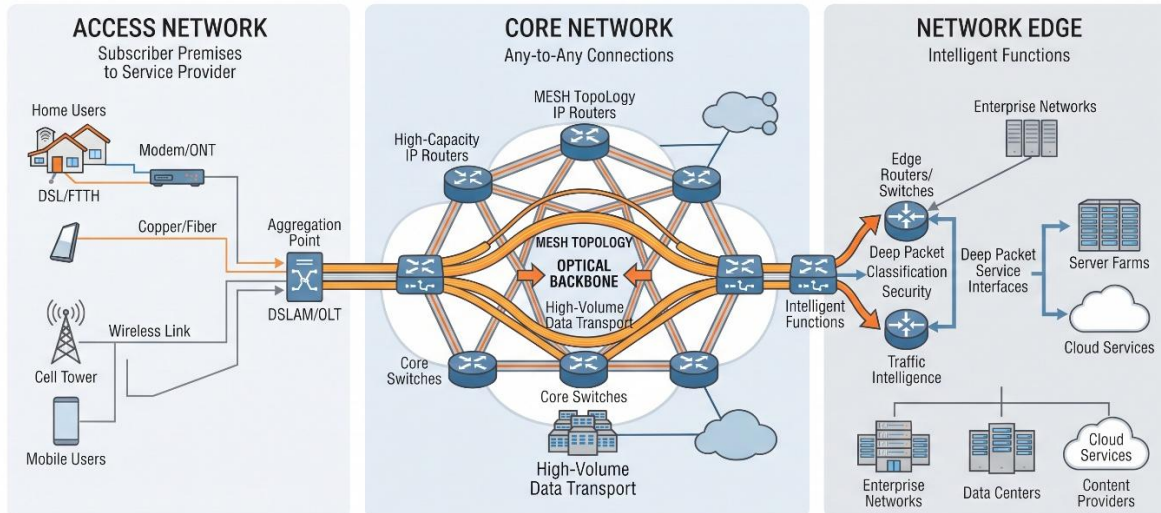


Network Infrastructures

Access Control

Modern communication networks are structured into distinct functional layers that work together to transport data from end-users to global destinations. Understanding this architecture requires examining three primary components: the **Access Network**, the **Core Network**, and the **Network Edge**.



Access Network

The **access network** is the foundational segment of a communications infrastructure that **physically connects end-user devices to their immediate service provider** (also known as the Internet Service Provider or local carrier).

Historically, telecommunication carriers relied primarily on traditional copper lines (twisted-pair telephone wires) designed for analog voice telephony. Today, these same physical lines—along with hybrid coaxial-fiber setups—are utilized to deliver high-speed **broadband internet** alongside traditional telephony signals. Due to the exponential growth in bandwidth demand, carriers are heavily investing in **optical fiber** deployments, which offer high data rates, immunity to electromagnetic interference, and stable performance over long distances.

Core Networks

The **core network** functions as the **high-capacity backbone of the telecommunications infrastructure**. It is typically structured in a mesh topology to guarantee robust, any-to-any connectivity (? A mesh layout provides multiple redundant paths between nodes. If one link or router fails, traffic can be dynamically rerouted through alternative paths, ensuring high availability and fault tolerance).

The core consists of high-performance IP routers and core switches interconnected by a high-speed optical backbone. By leveraging fiber optics at the core level, carriers can transport massive aggregates of data traffic across cities, countries, and continents with minimal latency and packet loss.

The Network Edge

The **network edge** refers to the **boundary where end systems (hosts, servers, enterprise networks) interface with the core network**. The edge may perform intelligent functions that are not performed inside the core network. In this scenario we divide:

- **Edge Devices (Switches/Routers):** These devices perform deep packet inspection or examine packet headers to classify traffic, apply security policies, shape traffic, and make initial routing decisions based on specific packet properties.

- **Core Devices:** Once the edge has processed and directed the traffic onto a chosen path, the core network focuses exclusively on high-speed, high-volume packet switching and forwarding without performing complex inspections.

This functional separation significantly enhances overall network performance. By keeping the core lightweight and focused entirely on rapid forwarding, bottlenecks are minimized, allowing the network to scale effectively to handle massive traffic volumes.

Types of Access

Wired: This category uses physical metallic or glass cables to establish a connection to the network. It guarantees high reliability, stable performance, and fast transmission speeds.

- **Powerline Access:** This technology uses existing electrical wiring within a building for network connectivity. It offers an easy setup without requiring new cables, making it useful for home networking and extending Wi-Fi coverage. However, it offers the least stable and lowest-performing connection quality among wired options.
- **DSL (Digital Subscriber Line) Access:** leverages traditional copper telephone lines to transmit digital data alongside voice signals. While constrained by distance from the telephone exchange, it offers the advantages of widespread availability and cost-effectiveness, making it historically popular for home internet and small businesses.
- **Cable Access:** Cable access utilizes coaxial cables—originally deployed for television broadcasting—to deliver bundled internet and TV services. It relies on a shared infrastructure among local subscribers to provide high-speed residential broadband.
- **Fiber-Optic Access:** Utilizing thin glass or plastic fibers to transmit data via light pulses, fiber-optic access delivers exceptional performance. Its primary advantages are high bandwidth, low latency, and high security, making it the standard for high-speed internet backbones and data centers.

Wireless: Wireless technologies use radio signals to connect devices to the network. Their key advantages include user mobility, architectural flexibility, and quick deployment. Typical examples range from local Wi-Fi networks to short-range Bluetooth connections.

- **Satellite Access:** connects remote terminals to communication satellites orbiting the Earth. Although it provides unmatched geographic coverage for remote and rural areas, traditional satellite links suffer from inherently high propagation latency (due to the vast distance to space) and environmental susceptibility.
- **Cellular Access:** Connecting via mobile network base stations, cellular access ensures continuous mobility, flexibility, and on-the-go connectivity. It is predominantly utilized by smartphones and mobile devices for wide-area cellular data, but struggle with bandwidth limitations and variable signal strength depending on user location and cell tower congestion.
- **5G Access:** Representing the fifth generation of mobile networks, 5G offers dramatically enhanced performance with ultra-high speeds and ultra-low latency. Beyond human-centric communications, it supports massive machine-type communications for the Internet of Things (IoT) and immersive technologies like Augmented/Virtual Reality (AR/VR), making it foundational for smart cities.

Wired Access

Copper Access

The copper access network was historically developed to support voice communications, but then evolved to meet both traditional telephone services and high-speed broadband.

To achieve this, operators implement digital subscriber line standards such as ADSL (Asymmetric Digital Subscriber Line) and VDSL (Very-high-bit-rate Digital Subscriber Line), which separate the transmission medium into discrete frequency bands. In addition, the major advantage of this network is its widespread availability and the use of existing infrastructure.

However, existing copper-based networks have gradually been expanded during several decades and their architectures are not optimized with regard to use of current technologies. If an entirely new network were to be built today, it would not be based on use of copper-based technologies (in favor of optical media), and the design would therefore be very different from those of today's copper-based networks.

One problem is that networks are designed mainly for carrying POTS, while a growing share of the traffic is based on IP or other data communication protocols, and in some areas, there are problems with capacity and quality of service.

Optical Access

Optical access systems deliver ultra-high speeds with high reliability, making them the primary choice for fixed access networks. Unlike copper, optical transmission eliminates distance-related attenuation and electromagnetic interference. The main technology types used depend on the deployment scenario:

- **GE-PON (Gigabit Ethernet Passive Optical Network):** mainly adopted for **individual** residential users.
- **G-PON (Gigabit Passive Optical Network):** typically deployed for **enterprise** and business customers.
- **CDM (Code Division Multiplexing over Fiber):** used to cover **long-distance** transmissions.

FTTx (Fiber-to-the-x)

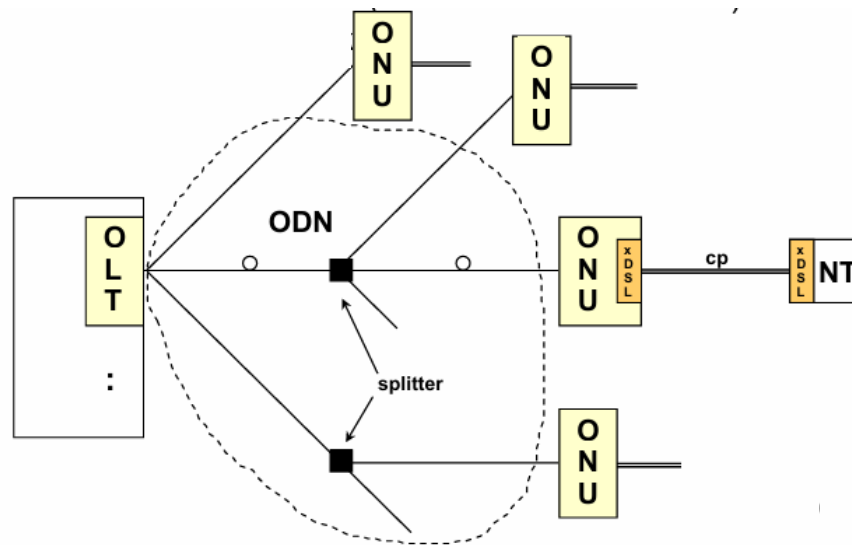
While technologies like PON define *how* data is transmitted, FTTx (Fiber-to-the-x) describes the physical deployment architecture—specifically, how close the optical fiber gets to the end user before switching to legacy copper wiring or terminating directly at the subscriber. These architectures replace traditional copper infrastructure by running fiber from a network element, such as the central office or local exchange, directly to a target destination denoted by x:

- **FTTH (Home):** The fiber runs all the way into the individual residence.
- **FTTP (Premise) / FTTU (User):** The fiber extends directly to the specific property or user.
- **FTTB (Building / Business):** The fiber reaches the boundary of the building (usually a basement or telecom room), while internal cabling handles distribution.
- **FTTO (Office) / FTTD (Desk):** The fiber reaches deep inside commercial environments, down to individual office rooms or desks.
- **FTTC (Curb):** The fiber reaches a street cabinet near the pavement/curb, with existing copper covering the final short distance.
- **FTTN (Node / Neighborhood) / FTTZ (Zone):** The fiber terminates at a shared distribution node serving a larger neighborhood or zone.

FTTx Elements

An FTTx deployment relies on four standard building blocks to transmit data between the service provider and the customer:

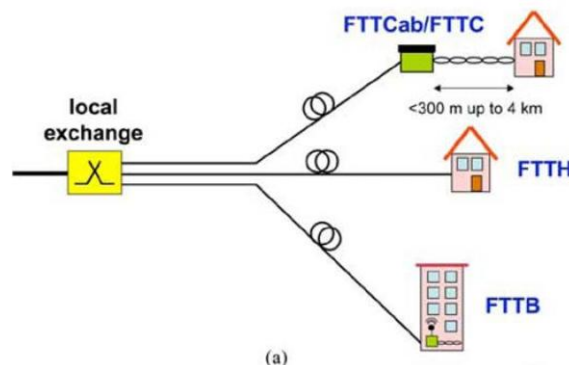
- **OLT (Optical Line Terminal):** The endpoint located at the service provider's Central Office (CO). It controls the bidirectional flow of information across the optical network and connects to the core IP network.
- **ODN (Optical Distribution Network):** The physical transmission path connecting the OLT to the ONUs, consisting of fiber cables, connectors, and splitters.
- **ONU (Optical Network Unit):** The device placed near or at customer premises that converts optical signals into electrical signals for end-user equipment.
- **ONT (Optical Network Termination):** A specific type of ONU deployed directly at the customer premises to serve as the local network termination point.



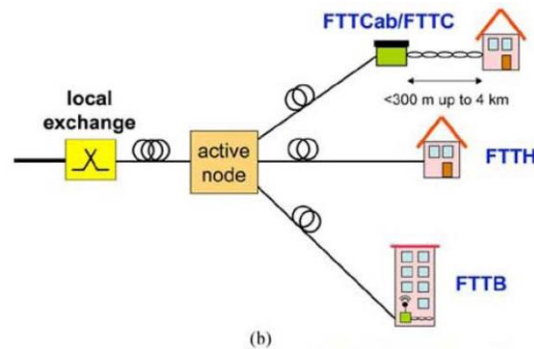
FTTx Architectures

FTTx networks are deployed using three main architectural models based on how the transmission medium is shared and managed:

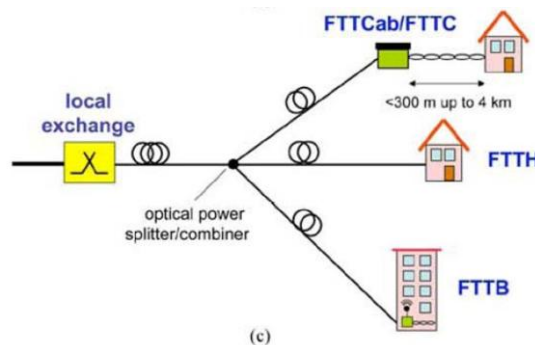
- **AON (Active Optical Network):** Also known as Point-to-Point (P2P), this setup provides a dedicated optical fiber directly from the central office switch to each individual subscriber, ensuring the highest network performance. However, this method costs too much to be really used in a normal scenario. As we can see in the picture (a) it requires a specific dig for each user (3 users => 3 digs) increasing the cost. So, this method costs so much due to the high cost of digging, (about 70 euros per meter, while the actual cost of fiber is extremely low, around 0.5 cents per meter.



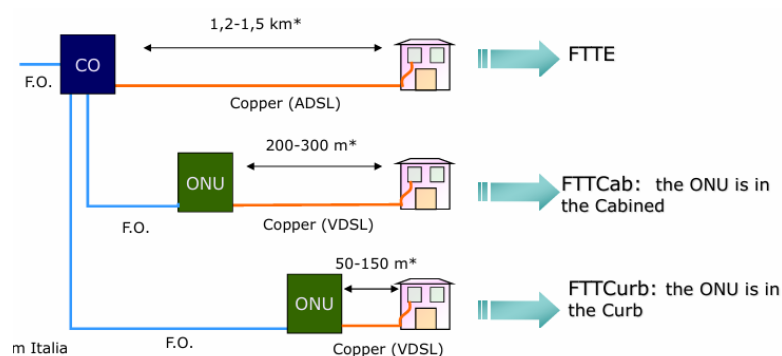
An intermediate solution (b) relies on electrically powered switching (the active node) that receives the light from the fiber, convert it into electricity, operates as a “router” and then retransmits light to the users. However, this method removes the advantage of dedicated fiber lines (the line from the local exchange to the active node it’s shared with the neighbourhood) and introduces operational drawbacks: the active nodes require electrical power, hardware maintenance, dedicated physical space, and represent potential single points of failure. Furthermore, dedicated runs increase deployment costs significantly (e.g., 3 endpoints require 3 separate fibers and potentially 3 trenching digs).



- PON (Passive Optical Network):** It’s the most used architecture. It’s a point-to-multipoint architecture based on a **tree topology**. Instead of dedicated fiber or powered intermediate switches, it uses unpowered **optical splitters** to passively divide a single optical signal from the OLT among multiple end users. This drastically reduces cabling requirements and maintenance costs since it doesn’t need electricity or space (no more active node, no switching, routing, no light->electricity->light, etc...). Practically, the splitter is a simple piece of fiber (usually made of glass) fused with the other fiber cables. In terms of performances modern PON standards deliver high speeds (from 1 Gbps up to 10 Gbps shared), easily meeting typical residential and business needs. However, individual performance can fluctuate slightly during peak traffic hours because users share the aggregate optical channel.



- Fiber to the Exchange:** A hybrid legacy model where optical fiber extends only as far as the Central Office (CO). The connection from the Central Office to the end customer is completed entirely over existing copper lines using digital subscriber technologies like ADSL. The primary disadvantage here is the copper local loop itself, which severely limits bandwidth, increases attenuation over distance, and suffers from electromagnetic interference. It’s the worst in term of performances: Signal attenuation increases sharply over distance, drastically reducing speeds as the subscriber gets farther from the exchange. Speeds are low and highly asymmetrical compared to all-optical solutions.



Wireless Access

The wireless access domain represents the sector holding the greatest promise for realizing the concept of ubiquitous networking. Its primary objective is to ensure communications without interruptions (*seamless*) and high-speed broadband connectivity by integrating both fixed and mobile access into a single architecture.

Talking about wireless access solution we must distinguish between:

- **Fixed Wireless Access:** instead of having cables interconnecting users and providers, FWA uses antennas. So, we have an antenna on the customer's building (e.g., a house or school) that communicates directly with the central provider antenna, located at a fixed base site.
- **Cellular:** A more complex infrastructure designed to provide connectivity to moving users over large geographic areas by distributing the coverage across adjacent radio cells.

Cellular Networks

Cellular networks are designed to provide wireless connectivity across **broad geographical regions** while actively managing **user mobility** and dedicated radio resources.

Even if these systems are named "Wireless Access" **the wireless segment is actually very small:**

- Only the radio hop between the end user and the base station is wireless.
- The rest of the infrastructure, connecting the base station antennas back to the core network, also known as the **backhaul** is completely wired.

Historically, the backhaul relied on copper lines (E1/T1 links up to 2G/3G), but it has since evolved to high-speed optical fiber to support growing data rates.

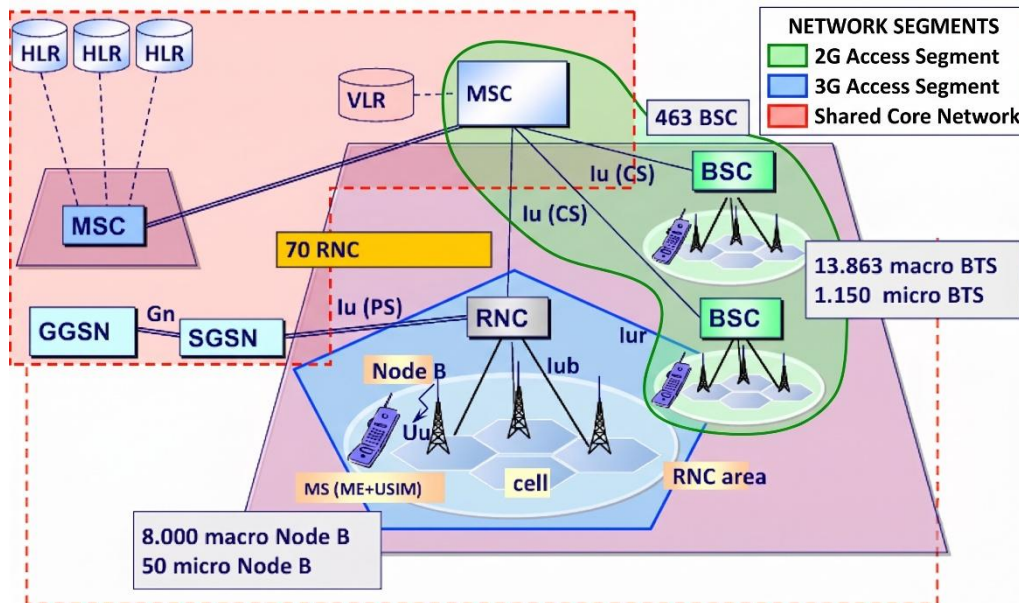
Characteristics

Cellular networks rely on two critical network operations:

1. **Dedicated Bandwidth Management:** Unlike Wi-Fi, which operates on a shared contention-based, in a Cellular Network the transmission resources (time, frequency, or codes) are allocated individually to each active user, ensuring strict quality of service and throughput management.
2. **Mobility Management:** Because cellular users can move freely (e.g., walking or traveling on a train), the network must support continuous connectivity through:
 - **Localization:** The network continuously tracks where the user terminal is located (Wi-Fi only associates a client with a local Access Point and does not know its wider location).
 - **Handover:** The connection must actively "follow" the user switching active data sessions from one antenna to another without dropping the service.

2G and 3G Architecture

In early cellular generations (2G and 3G, as well as 4G), the network divides the territory into geographical areas called **cells** (the name “Cellular” derives exactly from this. Cells => Cellular). Each cell is served by a Base Station (the physical antenna site).



- **2G Access Segment:** The mobile terminal communicates via radio with a base station known as a BTS (Base Transceiver Station). Multiple BTS units are aggregated and controlled by a hierarchical controller called a BSC (Base Station Controller).
- **3G Access Segment:** The user device (referred to as an MS - Mobile Station) connects via the Uu radio interface to the 3G base station, known as a Node B. Node Bs are managed and coordinated by an RNC (Radio Network Controller) via the Iub interface. Additionally, multiple RNCs can communicate directly with one another through the Iur interface to manage mobility and handovers.
- **Shared Core Network:** Both generations connect to a core network divided into two functional domains:
 - **Circuit-Switched (CS) Domain:** Dedicated primarily to traditional voice traffic, it centers on the MSC (Mobile Switching Center), supported by the HLR (Home Location Register) and VLR (Visitor Location Register) databases for location and user profiles. Both the BSC (2G) and the RNC (3G) interface with the MSC via dedicated links (the Iu(CS) interface for 3G).
 - **Packet-Switched (PS) Domain:** Dedicated to data traffic and Internet access; introduced with GPRS/EDGE and consolidated in 3G. The RNC connects via the Iu(PS) interface to the SGSN (Serving GPRS Support Node), which communicates via the internal Gn interface with the GGSN (Gateway GPRS Support Node) to route packets to external IP networks.

5G Architecture

In fifth-generation (5G) networks, the architecture is significantly simplified and modernized into a **Service-Based Architecture (SBA)** which clearly separates the control plane from the data transport plane. Like the older generations, the only wireless link remains the connection between the end user and the antenna, while the internal network interfaces (e.g., N2, N3, N4, N6) are high-speed wired links (usually fiber).

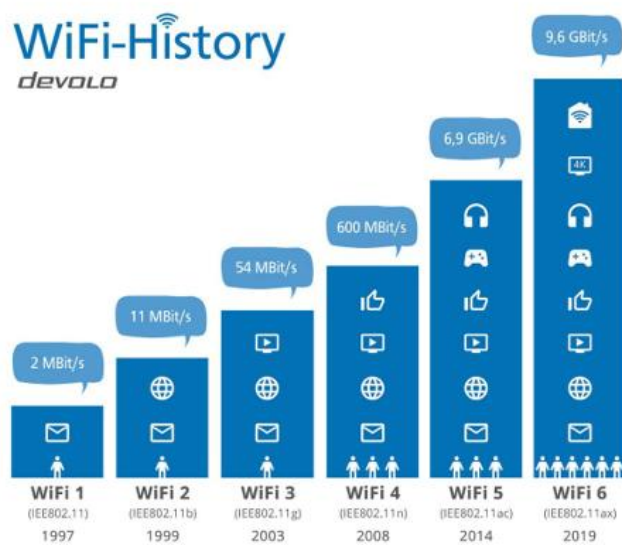


The key components include:

- **UE (User Equipment):** The end-user device (such as a smartphone, laptop, or IoT sensor) requesting network connectivity.
- **gNB (next-generation Node B):** The 5G base station antenna that manages the radio interface directly with the UE.
- **AMF (Access and Mobility Management Function):** responsible for authenticating the UE, handling network registration, and managing user mobility between gNB antennas.
- **SMF (Session Management Function):** A control entity that works alongside the AMF to allocate resources to each session.
- **UPF (User Plane Function):** It processes and routes data packets between the 5G radio access network and the external data network (Internet/Data Network). It's the part in which the packets are switched through the network to reach the end user.

Wi-Fi Evolution

The IEEE 802.11 family has progressively evolved over the past decades to address rising bandwidth demands, higher client densities, and media-rich services. To simplify technical nomenclature, the Wi-Fi Alliance retrospectively introduced user-friendly generation names (Wi-Fi 1 through Wi-Fi 6).



- **Wi-Fi 1 (IEEE 802.11 – 1997):** The legacy baseline standard offering a peak throughput of **2 Mbit/s**. It was primarily suited for basic, low-bandwidth text transmission like email.
- **Wi-Fi 2 (IEEE 802.11b – 1999):** Raised the maximum data rate to **11 Mbit/s** over the 2.4 GHz band. This standard enabled the early expansion of wireless home networking and general web browsing.
- **Wi-Fi 3 (IEEE 802.11g – 2003):** Introduced OFDM (Orthogonal Frequency-Division Multiplexing) in the 2.4 GHz band, increasing peak theoretical speed to **54 Mbit/s**. This allowed users to support standard video streaming along with browsing and email.
- **Wi-Fi 4 (IEEE 802.11n – 2008):** A major evolutionary step reaching up to **600 Mbit/s**. It introduced MIMO (Multiple-Input Multiple-Output) antenna technology and dual-band operation (2.4 GHz and 5 GHz), enabling multi-user environments, social media use, and higher-resolution video streaming.
- **Wi-Fi 5 (IEEE 802.11ac – 2014):** Focused heavily on the 5 GHz band, leveraging wider channels and MU-MIMO to push maximum throughput up to **6.9 Gbit/s**. This bandwidth headroom unlocked reliable online gaming, high-fidelity audio, and dense multi-device connectivity.
- **Wi-Fi 6 (IEEE 802.11ax – 2019):** Designed to deliver up to **9.6 Gbit/s**. Beyond raw speed, its main improvement is spectral efficiency in crowded environments using OFDMA (Orthogonal Frequency-Division Multiple Access), making it capable of supporting 4K/8K media streams and high client densities simultaneously. It also allows us to make calls on the mobile network (e.g., if you are abroad and have no roaming active).

GEO Satellites Access

GEO (Geostationary Orbit) is a type of access that relies on satellites which orbit at an altitude of approximately **35,786 km** directly above the Earth's equator. At this specific distance, their orbital speed matches the rotation of the Earth, which means a GEO satellite appears completely stationary in the sky from the perspective of a ground observer.

Because they are positioned so high, just **three satellites** are mathematically sufficient to provide near-global coverage and ground antennas do not need to track them across the sky (since the satellites seem stationary from Earth, the antennas simply “point at a fixed location in the sky”).

However, this massive distance introduces an inevitable physical delay with a transmission latency of **500 ms or more**, making GEO unusable for today's real-time interactive traffic (like voice calls, online gaming, or rapid two-way communication).

LEO Satellites Access

LEO (Low Earth Orbit) solves the latency problems of GEO by relying on much closer satellites, which orbit at an altitude of approximately **500 to 2000 km** above Earth (15-20 times closer than GEO).

However, due to their proximity to the surface, they have a short orbital period of roughly **90 to 120 minutes** per orbit, meaning a single satellite cannot provide persistent local service. To solve this problem and guarantee continuous global coverage LEO requires **large satellite constellations** and specific algorithms that move clients from a satellite to another without connection loss.

- **Advantages:**

- **Low Latency:** Round-trip transmission time is roughly **20–40 ms**, which is comparable to terrestrial broadband and drastically lower than legacy satellite links.
- **High Bandwidth Potential:** Proximity enables strong signal reception and higher throughput capacity.
- **Global Reach:** Delivers reliable coverage to remote, rural, and maritime areas underserved by fiber or mobile networks.

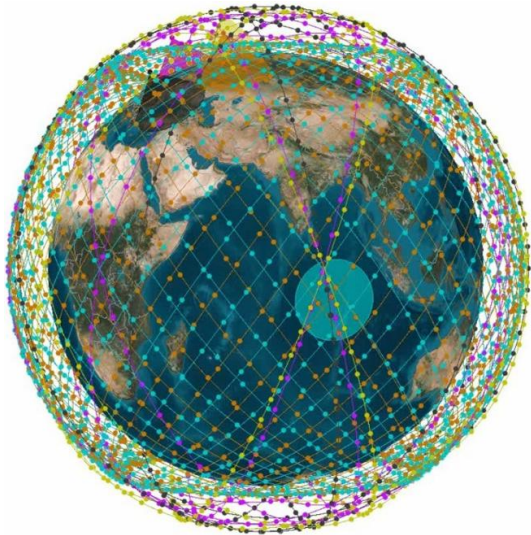
- **Challenges:**

- **Constellation Size:** Thousands of active satellites are required to ensure continuous handovers without dropped connections.
- **High Deployment and Maintenance Costs:** Launching, managing, and replenishing a massive constellation involves major capital investment.
- **Short Lifespan and Space Debris:** Operating in low orbit subjects spacecraft to atmospheric drag, resulting in a short operational lifespan of only **5 to 7 years** and increasing orbital debris risks.

LEO vs. GEO

Feature	LEO (Low Earth Orbit)	GEO (Geostationary Orbit)
Altitude	~500–2,000 km	~35,786 km
Latency	Low (~20–40 ms)	High (~500+ ms)
Coverage Mechanism	Requires thousands of satellites in a constellation	A single satellite covers roughly 1/3 of the Earth
Lifespan	Shorter (~5–7 years)	Long (~15 years)

Example: Starlink



Developed by SpaceX, **Starlink** represents the most prominent commercial deployment of a broadband LEO mega-constellation:

- **Scale and Purpose:** Over 5,000 satellites deployed to deliver global broadband coverage to residential users, enterprises, and mobility sectors (such as aircraft and maritime vessels).
- **Terminal and Performance:** End users connect via a compact, phased-array antenna terminal ("Dishy"), achieving typical round-trip latencies of **20–40 ms**.

Backbone Networks

A telecommunications backbone is split into two distinct architectural views:

- **Logical Topology:** which describes how IP traffic and routing protocols see the network).
- **Physical Topology:** the underlying level that actually contains fiber optics and transmission equipment in the ground.

Logical Topology: The Optical Packet Backbone (OPB)

The Optical Packet Backbone (OPB) represents the packet-switched routing layer consisting of 32 Points of Presence (PoPs) distributed across the country. It is organized in a hierarchical, two-tier star/mesh structure to aggregate and forward IP traffic efficiently:

- **Inner Core (4 PoPs):** The central transit backbone consists of 4 high-capacity PoPs: 2 located in Rome and 2 in Milan. These main nodes are interconnected with heavy trunk links operating at 10 Gbit/s (STM-64).
- **Outer Core (28 PoPs):** The remaining 28 regional PoPs aggregate local traffic and connect back to the Inner Core. Depending on local traffic demand, these links utilize different transmission rates:
 - 10 Gbit/s (STM-64) for high-volume secondary nodes.
 - 2.5 Gbit/s (STM-16) for intermediate regional nodes.
 - 155 Mbit/s (STM-1) for lower-demand peripheral links.
- **Dimensioning Rule (50% Link Utilization):** Core links are intentionally engineered to operate at an average load of 50% capacity. This ensures adequate headroom to absorb traffic spikes and allows immediate failover rerouting without congestion if an alternate link goes down.

Physical Topology: The Optical Transport Network (OTN)

The logical OPB does not run on dedicated end-to-end dark fiber for every single logical link; instead, it is built directly on top of an underlying physical Optical Transport Network (OTN).

- **Physical Nodes (ODXC):** Physical transmission sites house ODXC (Optical Digital Cross-Connect) equipment. These devices switch high-capacity lightpaths and groom optical wavelengths directly at the physical layer without needing packet inspection.
- **Multiplexed Transmission (DWDM):** Long-distance physical spans use DWDM (Dense Wavelength Division Multiplexing) over optical fiber. DWDM allows many individual optical channels (lambdas) to travel down a single physical pair of fibers simultaneously.
- **Logical to Physical Independence:** The logical map looks like a direct, simplified tree/star radiating from Rome and Milan. However, the physical map reveals a heavily meshed ring topology connecting physical cities (e.g., Turin, Genoa, Florence, Bologna, Naples, Palermo). If a physical optical link is severed, the DWDM/ODXC layer can automatically reroute the optical channel through an alternate geographic path while the logical IP layer above remains completely unaware of the physical fault.